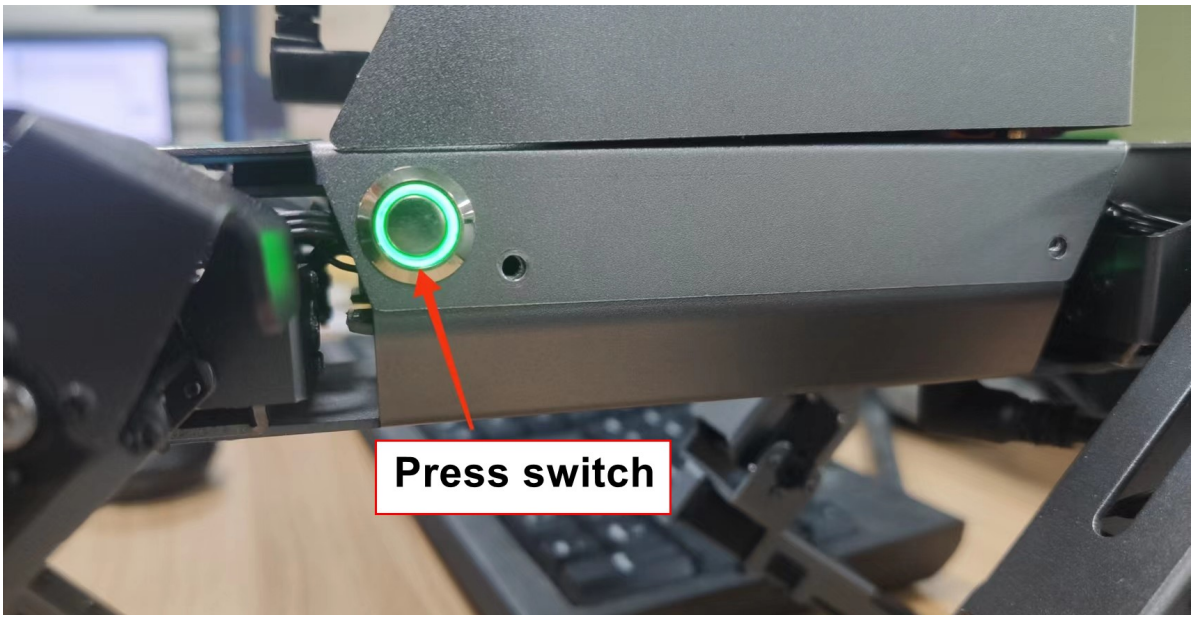
ROS2 environment entity robot dog navigation

Quick use

1. Power on the robot dog

PI4 version steps:

Press the power switch on the side of the robot dog and wait for the robot dog to start up. And connect the robot dog's WiFi to the same LAN environment as the computer.



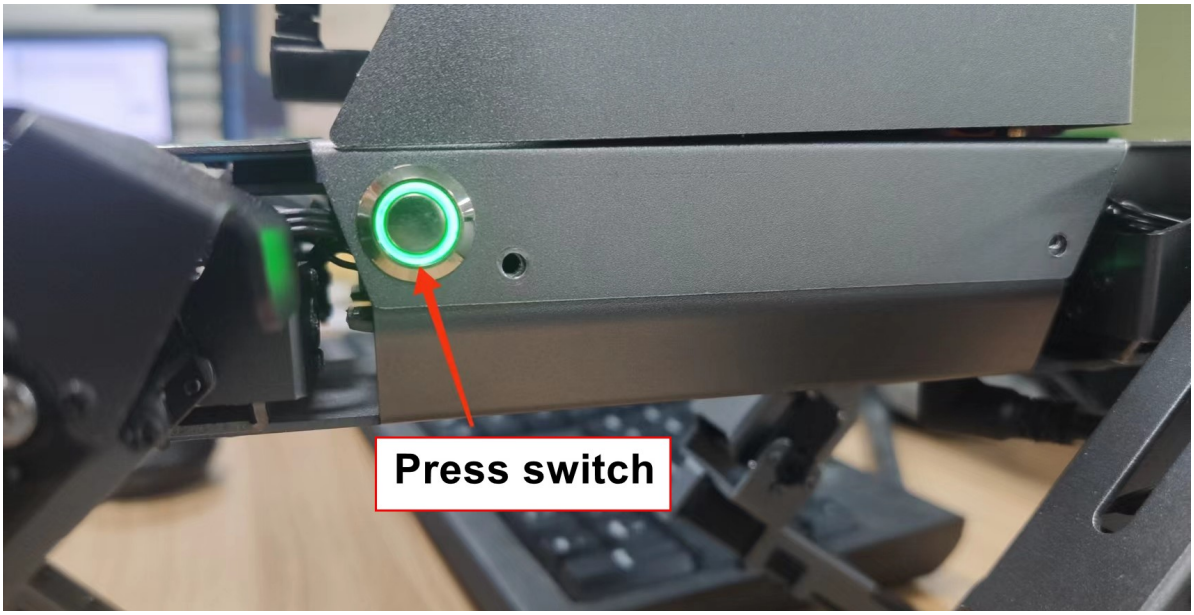
After the robot dog is started, the laser radar, imu, and robot dog joint status nodes have been automatically started.

If you find that you cannot obtain lidar and other data, please close the robot dog program and restart the chassis program.

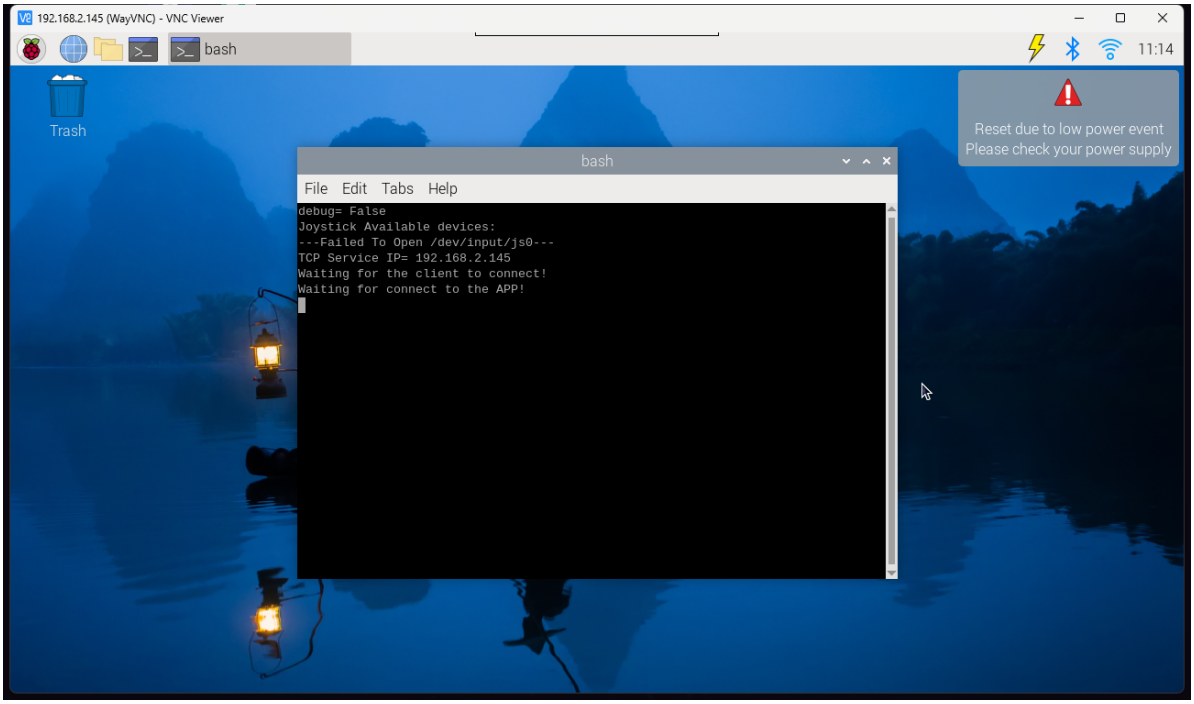
To close the large program, restart the chassis and modify the multi-level communication ID, please refer to the tutorial: 14. Radar mapping navigation\6. Obtaining the status of the physical robot dog in the ROS2 environment\Acquiring the real joint data of the robot dog in the ROS2 environment.pdf

PI5 version steps:

Press the power switch on the side of the robot dog and wait for the robot dog to start up. And connect the robot dog's WiFi to the same LAN environment as the computer.



After the robot dog is started, remotely connect to the robot dog through the IP address on the OLED.



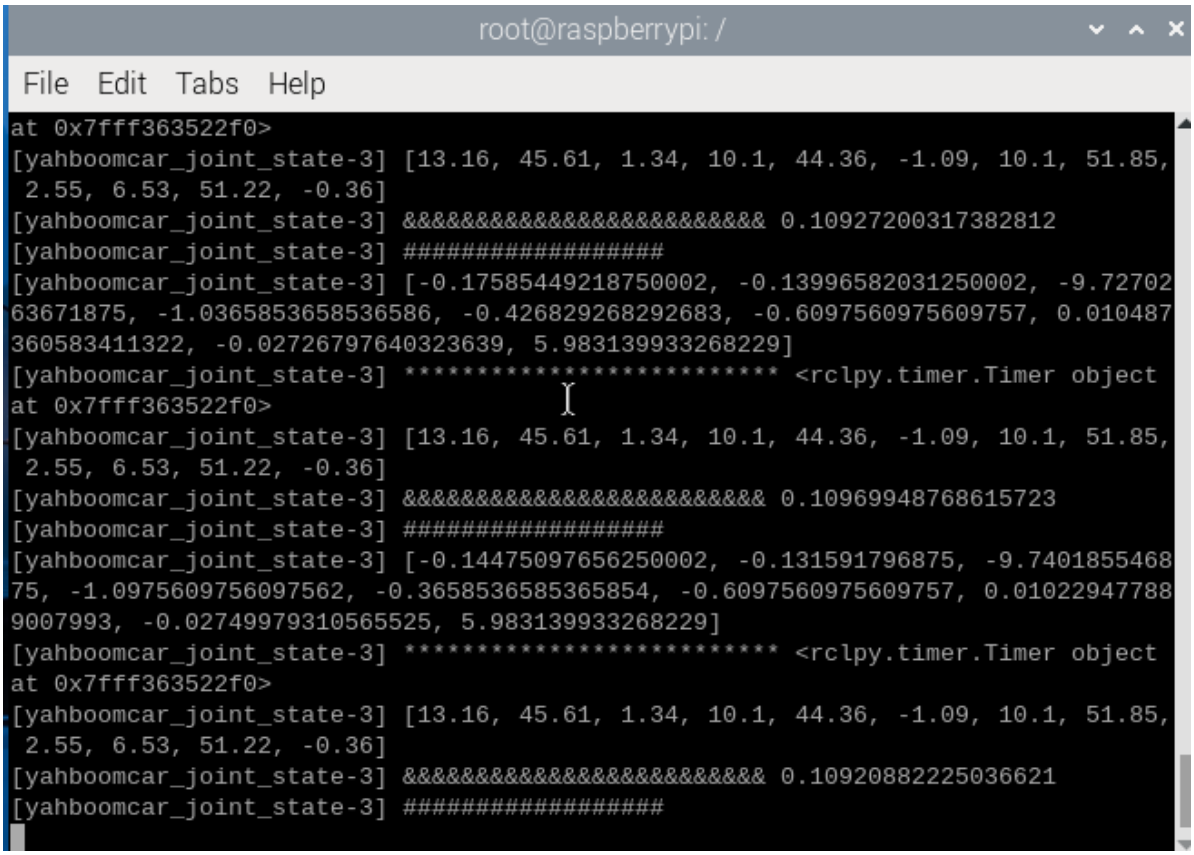
Then ctrl+c closes the large program and enter the following command to enter docker:

```
./run_humble.sh
```

```
TCP Service IP= 192.168.2.145
Waiting for the client to connect!
Waiting for connect to the APP!
^CKeyboardInterrupt
2024-04-28T10:17:27Z
-----program end-----
pi@raspberrypi:~ $ ./run_humble.sh
access control disabled, clients can connect from any host
root@raspberrypi:/#
```

Then enter the following commands in the docker terminal to start the car radar, imu, and robot dog joint status nodes.

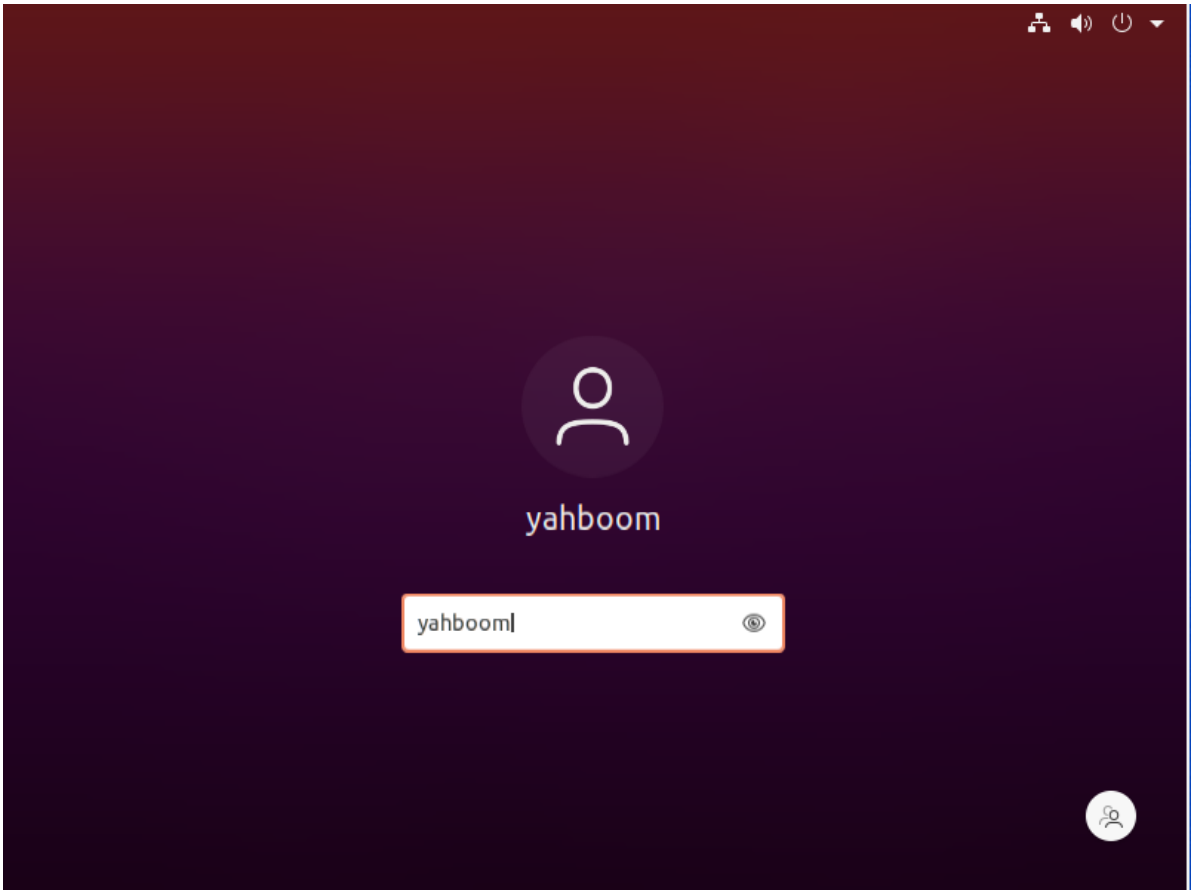
```
ros2 launch bringup Navigation_bringup.launch.py
```



2. Open the virtual machine

Note: The default virtual machine has been installed here.

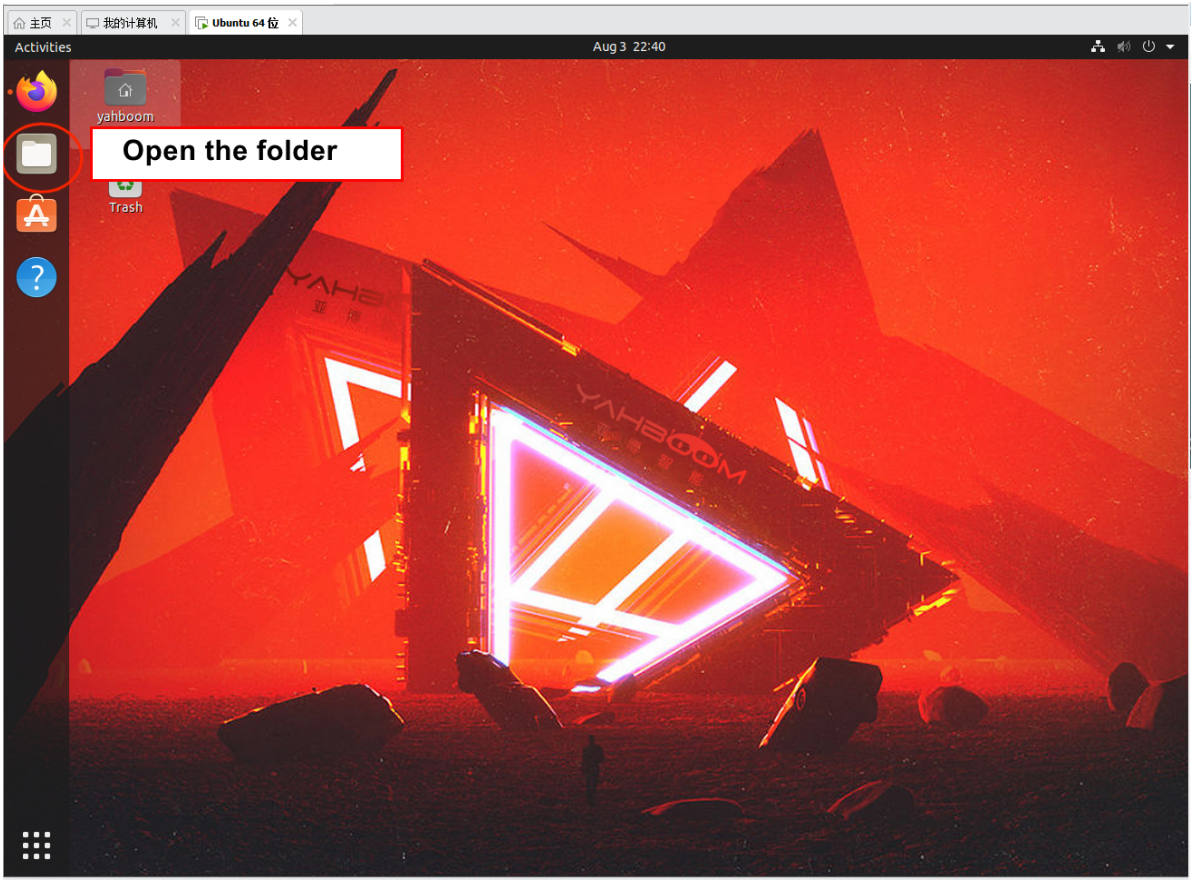
Open the virtual machine, enter the password: yahboom and press the Enter key to enter the system desktop.



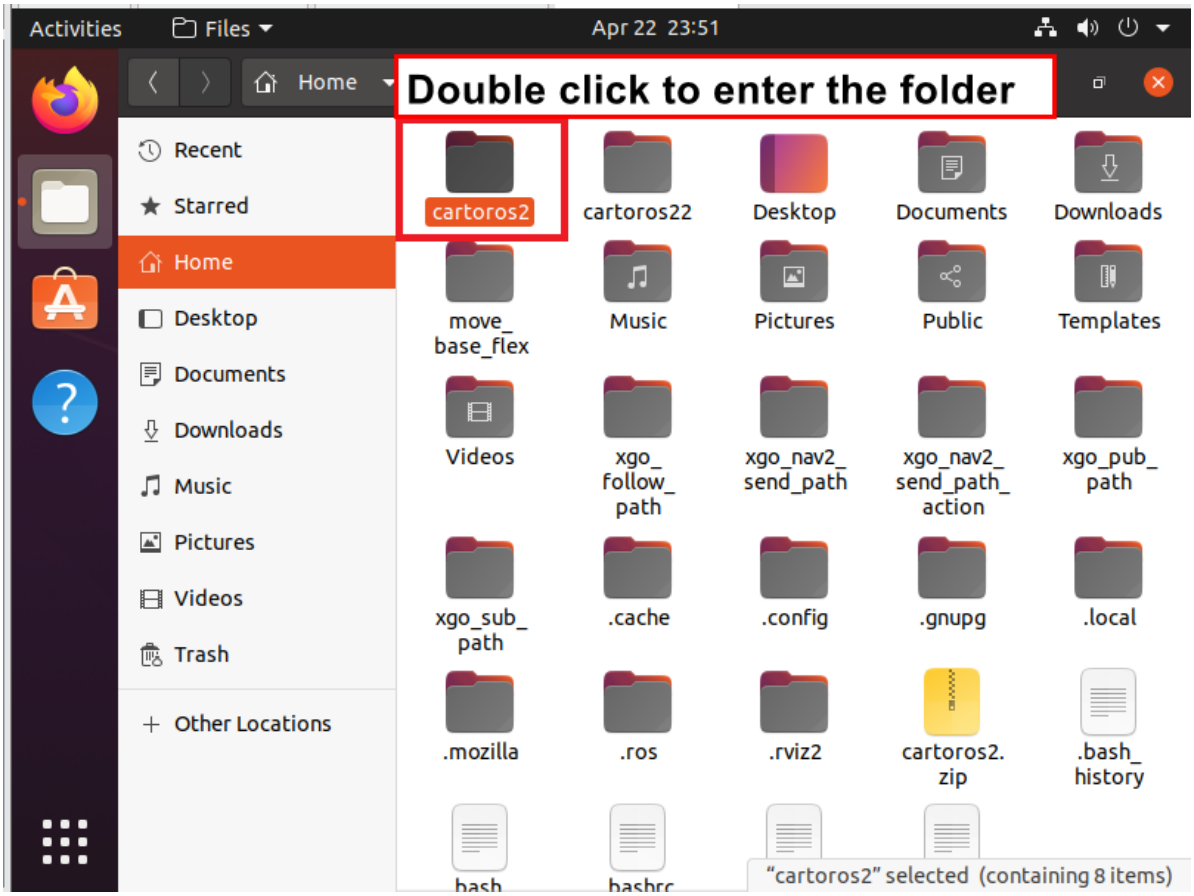
3. Start the robot dog navigation node

PI4 version steps:

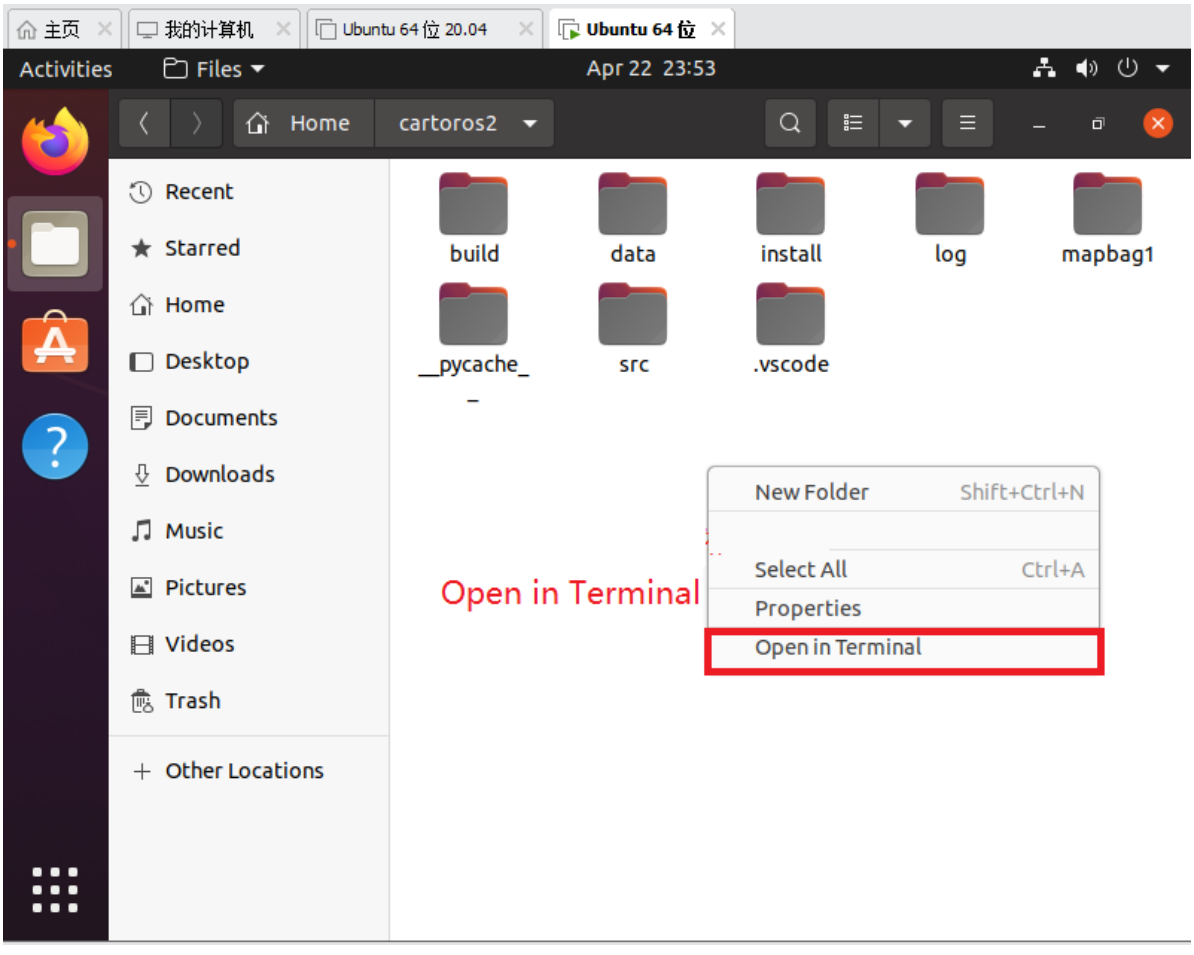
Enter the desktop system and open the folder.



Then double click on the cartoros2 folder



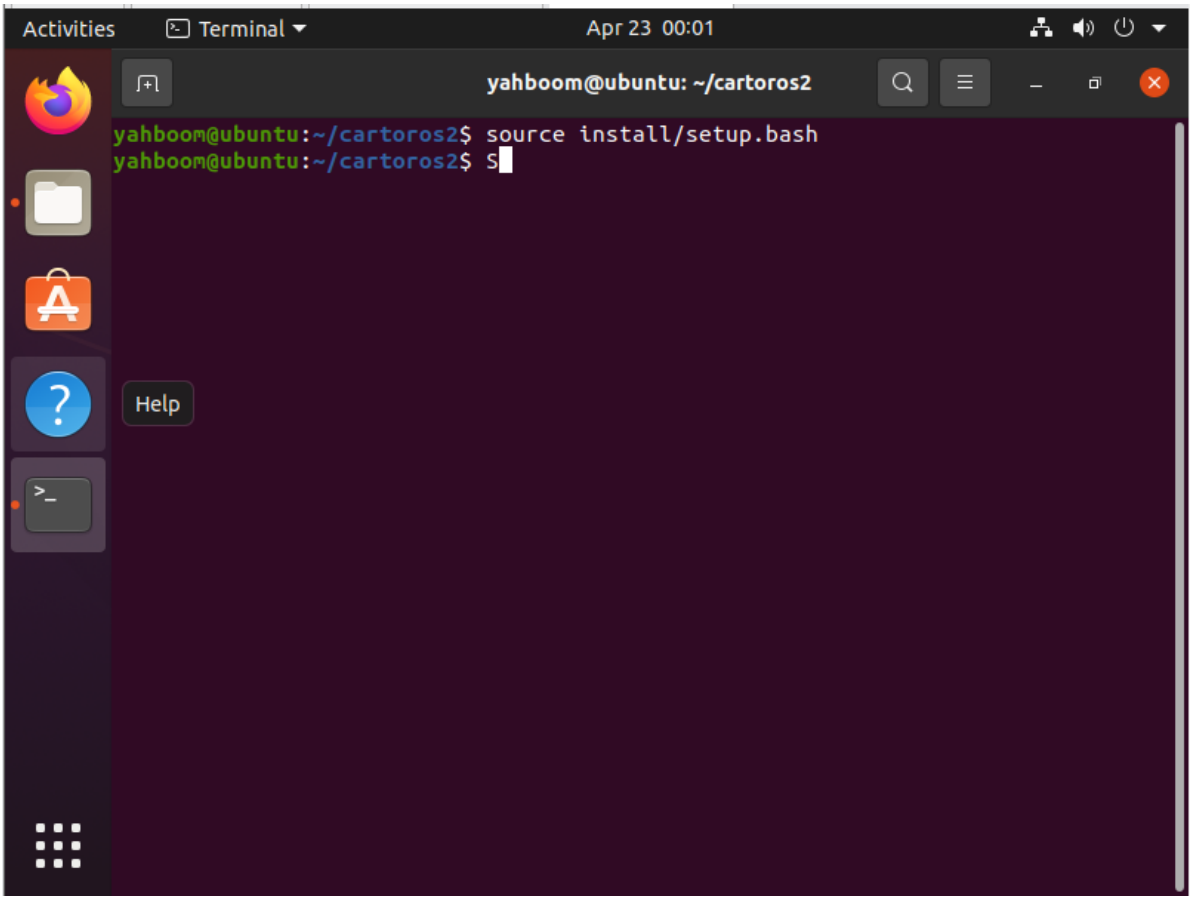
Then right-click in an empty space of the folder and select Open in Terminal



Then enter the following command in the terminal to activate the environment

```
source install/setup.bash
```

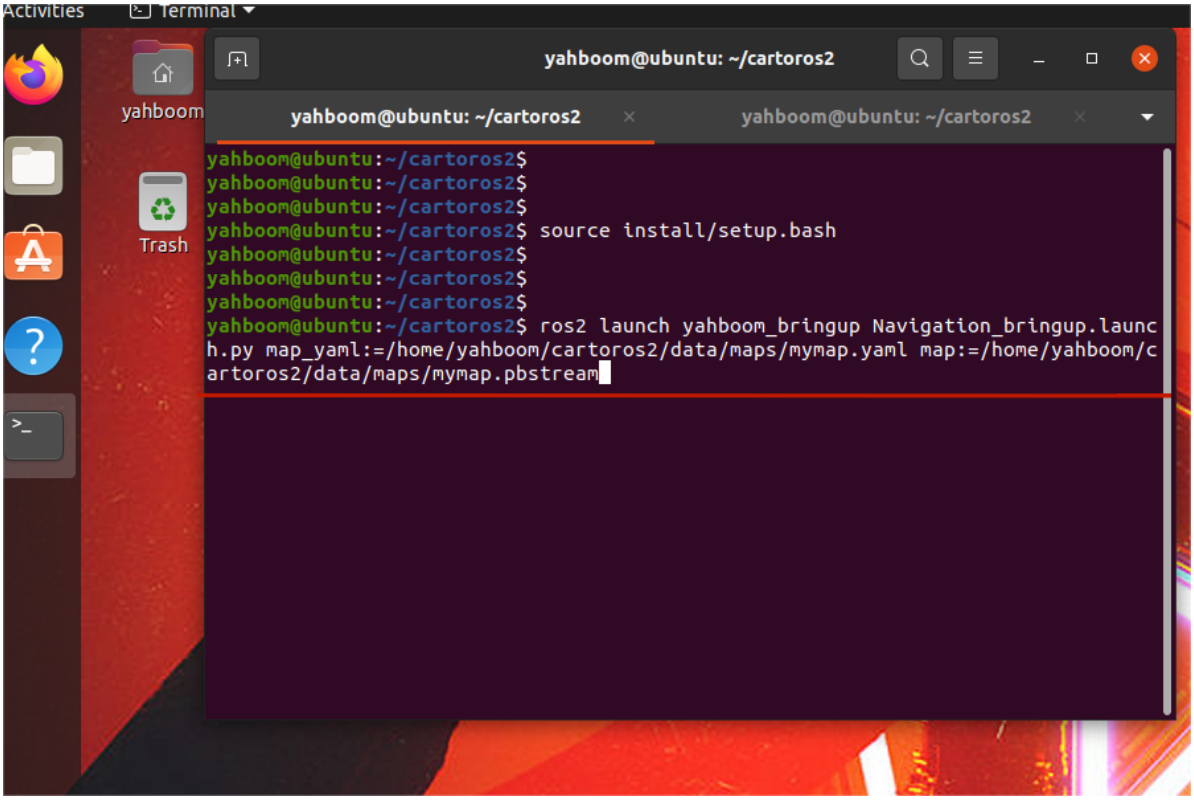
After completing the input, press the Enter key.



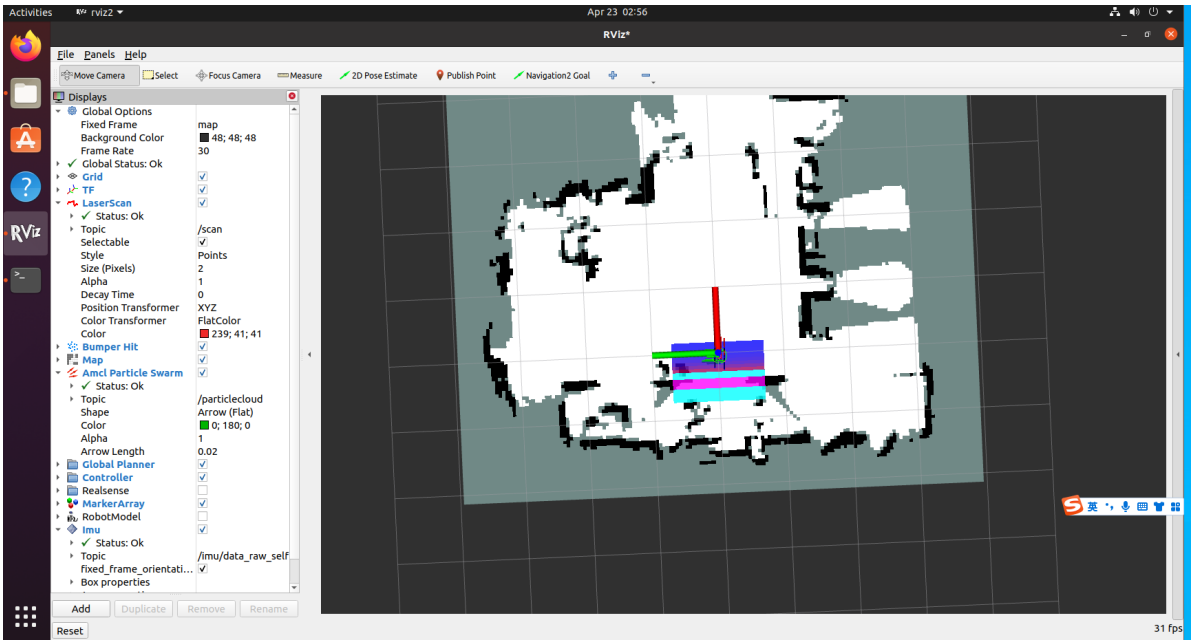
Then enter the command

```
ros2 launch yahboom_bringup Navigation_bringup.launch.py
map_yaml:=/home/yahboom/cartoros2/data/maps/mymap.yaml
map:=/home/yahboom/cartoros2/data/maps/mymap.pbstream
```

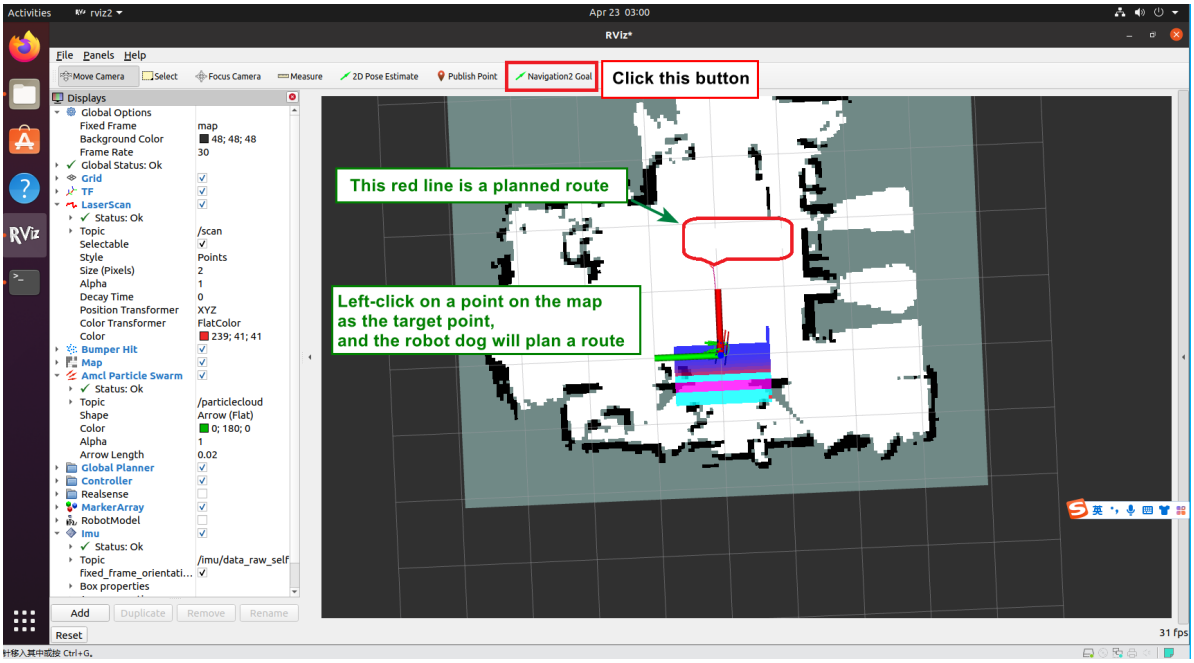
Where map_yaml is the yaml file of the map, and map is the pbstream file of the map.



Then press the Enter key to navigate.

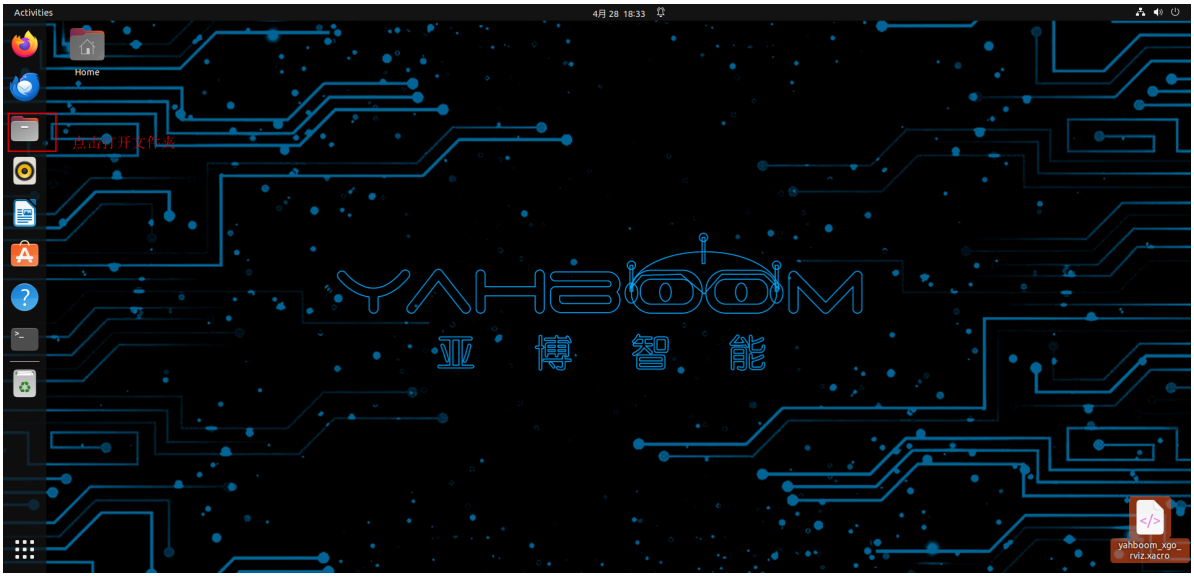


After clicking the Navigation2 Goal button, click a target point on the map interface, and the route will be automatically planned and driven to the target point.

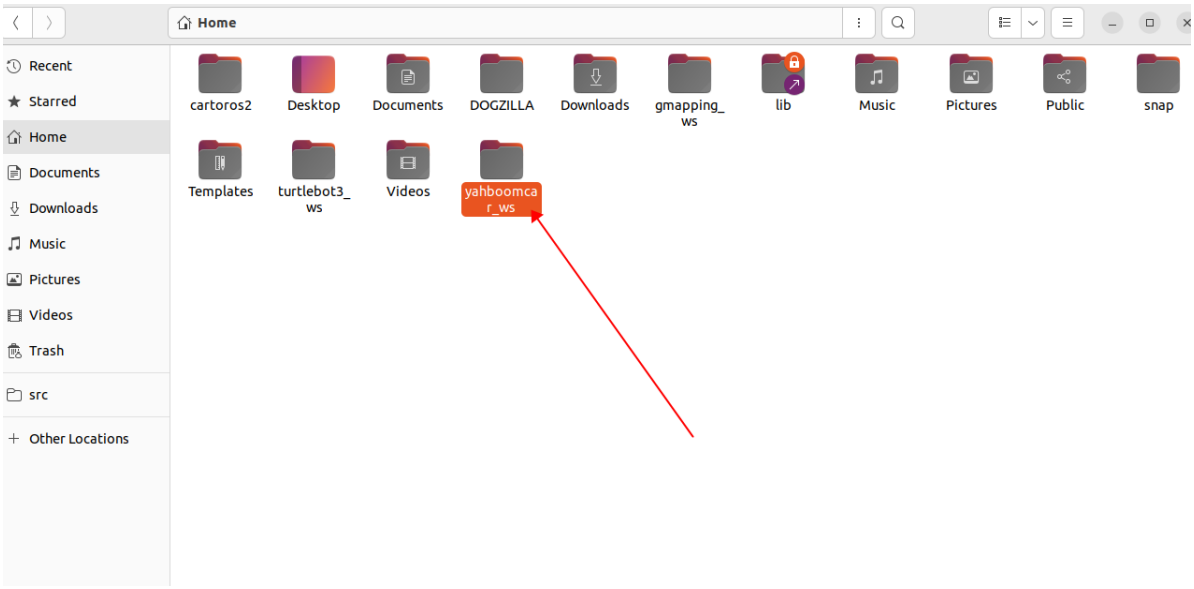


PI5 version steps:

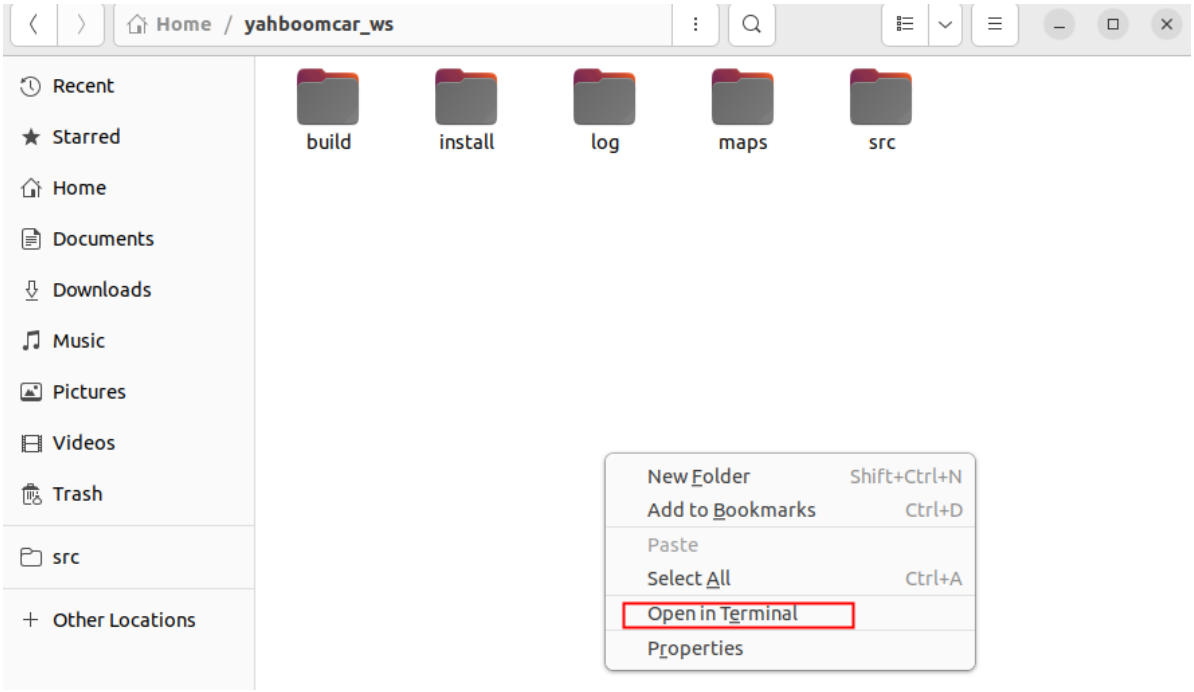
Enter the desktop system and open the folder.



Double-click to open the yahboomcar_ws folder



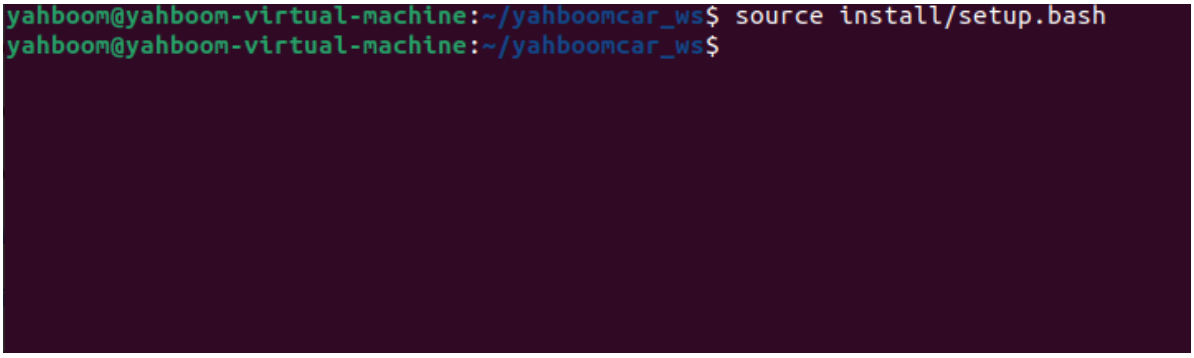
Then right-click in an empty space of the folder and select Open in Terminal



Then enter the following command in the terminal to activate the environment

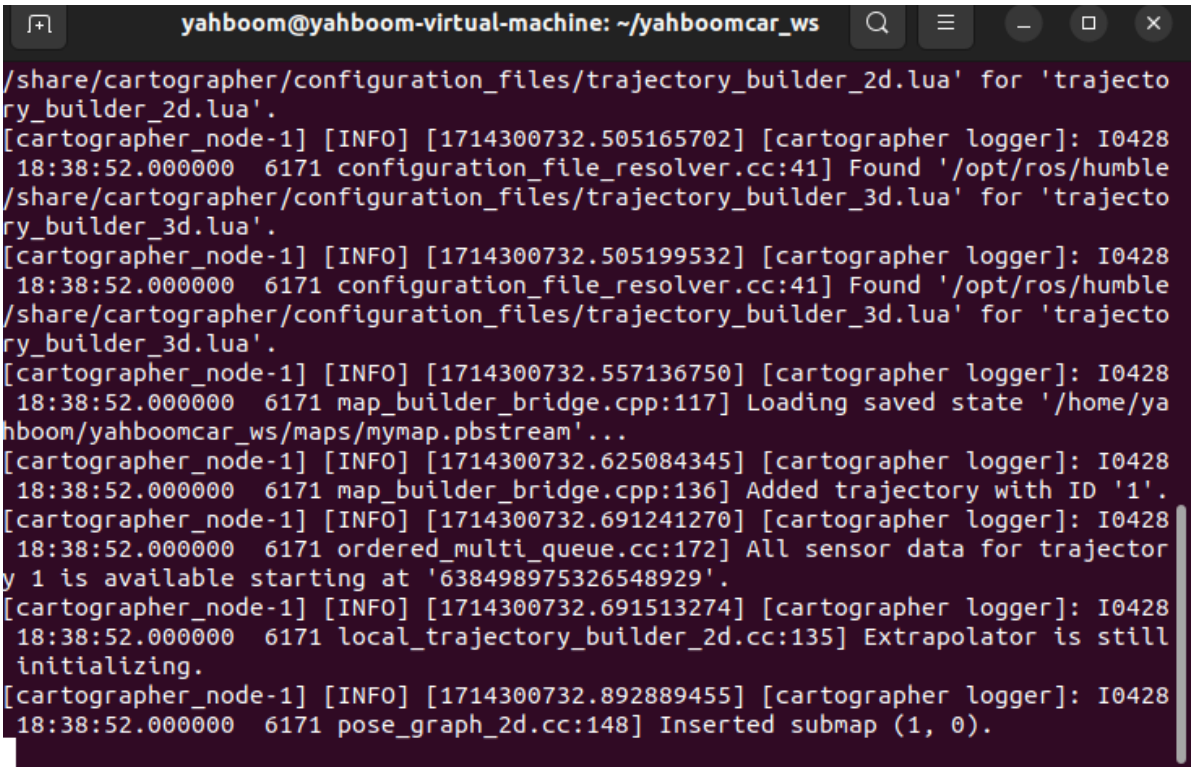
```
source install/setup.bash
```

After completing the input, press Enter.



Then enter the command to start the relocation function

```
ros2 launch yahboom_dog_cartographer localization_imu_odom.launch.py
load_state_filename:=/home/yahboom/yahboomcar_ws/maps/mymap.pbstream
```

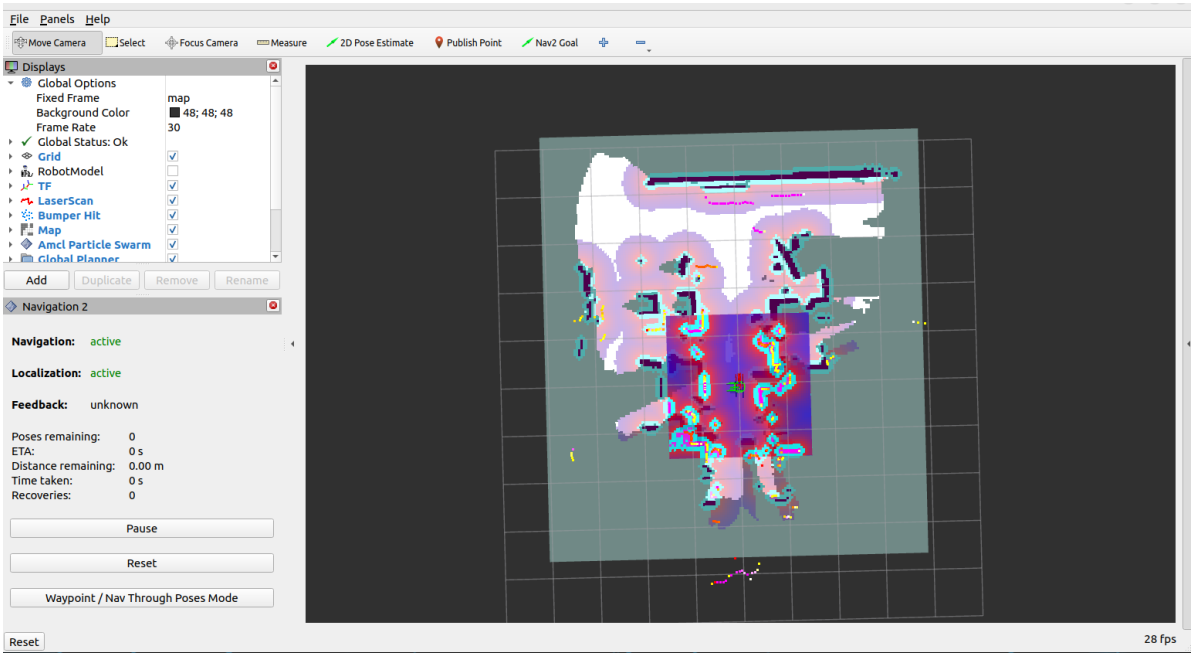


Repeat the above steps to open the terminal, reopen a terminal and enter navigation commands.

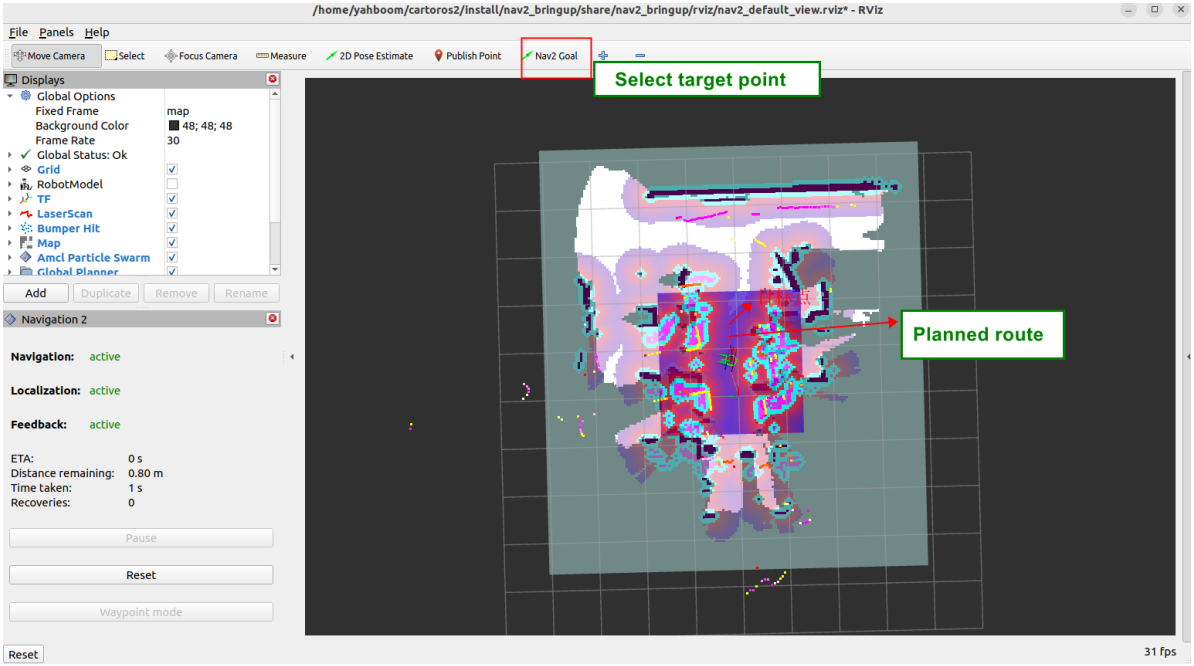
```
ros2 launch yahboom_dog_navigation2 navigation2.launch.py use_sim_time:=False
map:=/home/yahboom/yahboomcar_ws/maps/mymap.yaml
```

```
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ ros2 launch yahboom_dog_navigat
ion2 navigation2.launch.py use_sim_time:=False map:=/home/yahboom/yahboomcar_ws/
maps/mymap.yaml
```

Then press the Enter key to navigate.



After clicking the Navigation2 Goal button, click a target point on the map interface, and the route will be automatically planned and driven to the target point.


/home/yahboom/cartoros2/install/nav2_bringup/share/nav2_bringup/rviz/nav2_default_view.rviz* - RViz
File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
Publish Point
Nav2 Goal
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Grid
RobotModel
TF
LaserScan
Bumper Hit
Map
Amcl Particle Swarm
Global Planner
Add
Duplicate
Remove
Rename
Navigation 2
Navigation: active
Localization: active
Feedback: active
ETA: 0 s
Distance remaining: 0.80 m
Time taken: 1 s
Recoveries: 0
Pause
Reset
Waypoint mode
Reset
Select target point
Planned route
31 fps